Evaluating University Selection Criteria and Applicant Decision-Making in College Admissions using TOPSIS

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Abstract

The college admissions process is a high-stakes decision for both prospective students and the institutions evaluating them. An estimated 60% of graduating students rank college admissions as their most stressful academic experience, a stressor traceable in part to the lack of objective decision support (Li et al., 2019). On the institutional side, manual review processes routinely fail to balance objectivity against institutional priorities, which can misalign admitted classes with intended program outcomes (Sucipto and Wibisono, 2019). This paper reviews the literature on multi-criteria decision-making (MCDM) for higher-education applications, with primary focus on the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) and its fuzzy extension (FTOPSIS), often paired with the Analytic Hierarchy Process (AHP) for criterion weighting. Across seven empirical case studies, with country settings spanning Turkey, Vietnam, India, Indonesia, and Nigeria, TOPSIS-based decision support systems consistently produce reproducible rankings, accommodate criteria sets ranging from 5 to 40 sub-metrics, and—where compared directly to manual review—surface divergences that the source literature interprets as corrections to subjective committee judgment rather than errors. The review concludes that TOPSIS–AHP pipelines are mature enough for institutional deployment, that FTOPSIS extends the framework usefully when input data is linguistic or imprecise, and that the principal remaining gap is integration with machine-learning techniques that could automate criterion-weight elicitation.

1 Introduction

1.1 Background and Motivation

The college admissions process is a dually critical gateway for academic institutions and prospective students. For students, the choice of where to enroll is a life-changing decision constrained by financial aid, location, program fit, and career outcomes; for institutions, the choice of whom to admit determines graduation rates, alumni outcomes, accreditation, and rankings. Both sides operate under uncertainty, and both routinely rely on heuristics rather than structured decision support.

The absence of standardized decision support systems (DSS) creates two distinct problems. First, application review is slow: committee-based evaluation cannot easily scale to the volume of applications most institutions now receive. Second, evaluation is non-uniform: different reviewers weight criteria differently, and even individual reviewers exhibit inconsistency across applications (Sucipto and Wibisono, 2019). The result is a process that is simultaneously expensive (in committee hours) and noisy (in outcomes), with high-achieving but non-traditional candidates being especially likely to be misranked.

Multi-criteria decision-making (MCDM) methods address this gap directly. Among MCDM techniques, the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) is widely used in higher-education contexts because it produces interpretable rankings, accommodates heterogeneous criteria, and allows uncertainty to be modeled explicitly through fuzzy extensions. This paper reviews the empirical literature on TOPSIS in college admissions and university selection, with attention to the criteria most frequently weighted, the modeling extensions used to handle imprecise data, and the gaps that remain.

1.2 Method Overview: TOPSIS

TOPSIS ranks alternatives by their geometric distance from two reference points: a positive ideal solution (the best score on every criterion) and a negative ideal solution (the worst score on every criterion). The alternative closest to the positive ideal and farthest from the negative ideal is preferred. The procedure has four steps: (i) normalize the decision matrix, (ii) apply criterion weights (often elicited via the Analytic Hierarchy Process), (iii) identify the positive and negative ideal solutions, and (iv) compute relative closeness coefficients for each alternative.

The normalization step uses linear normalization to map each decision-matrix entry x_{ij} (criterion j for alternative i) to a comparable scale:

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{k=1}^{m} x_{kj}^2}}, \quad i = 1, 2, \ldots, m, \quad j = 1, 2, \ldots, n. \tag{1}$$

The positive and negative ideal solutions S^+ and S^- are then constructed from the column-wise extrema of the weighted normalized matrix. Following Bagi et al. (2020), these are written as

$$S^+ = (y_1^+, y_2^+, \ldots, y_n^+), \qquad S^- = (y_1^-, y_2^-, \ldots, y_n^-), \tag{2}$$

where for benefit criteria $y_j^+ = \max_i h_{ij}$ and $y_j^- = \min_i h_{ij}$, and the inequalities are reversed for cost criteria. The Euclidean distances of each alternative to the two ideals, and the resulting closeness coefficient R, are

$$D_i^+ = \sqrt{\sum_{j=1}^{n} w_j^2 \, (h_{ij} - h_j^+)^2}, \quad D_i^- = \sqrt{\sum_{j=1}^{n} w_j^2 \, (h_{ij} - h_j^-)^2}, \quad R_i = \frac{D_i^-}{D_i^- + D_i^+}. \qquad (3)$$

Alternatives are ranked in descending order of $R_i \in [0, 1]$, with R_i closer to 1 indicating better performance. Criterion weights w_j are typically elicited from domain experts via AHP pairwise comparisons or, in entropy-TOPSIS variants, derived from the dispersion of the data itself.

1.3 Scope and Structure

This review is organized around three questions. (1) How is TOPSIS applied on the institutional side, where universities rank applicants or evaluate themselves against peers? (Section 2.) (2) How is TOPSIS applied on the student side, where applicants rank universities or post-graduation career options? (Section 3.) (3) How is uncertainty in the input data handled, particularly when criteria are linguistic rather than numerical? (Section 4.)

2 Institutional-Side Applications

2.1 University Self-Evaluation: Istanbul Case

Gülsün and Miç (2019) applied TOPSIS to rank universities in Istanbul, Turkey, using survey responses from 100 students. Six criteria were weighted, with prestige, scholarship opportunities, and accessibility identified as the three most heavily emphasized in the source's discussion. The authors used AHP to elicit pairwise importance comparisons, computed the resulting normalized weight vector, and applied standard TOPSIS to rank the universities. The reported Consistency Index of 0.961 indicates that the AHP comparisons were internally coherent, and Yildiz University emerged as the top-ranked institution on the combined score. The study illustrates a now-typical TOPSIS–AHP pairing: AHP handles the subjective weight elicitation, TOPSIS handles the ranking once weights are fixed.

2.2 University Self-Evaluation: Vietnam Case

Wang et al. (2022) extended this approach to 45 private universities in Vietnam using an entropy-TOPSIS variant that bypasses AHP and instead derives criterion weights from the dispersion of the underlying data. Seven criteria were used: two staff performance measures (C_1, C_2), training facilities (C_3), two enrollment-type measures (C_4, C_5), placement rates (C_6), and tuition revenue (C_7). Hutech University ranked first with a closeness coefficient of 0.7557 averaged over two years, and the analysis flagged systematic gaps in teacher performance and facility availability across the lower-ranked institutions. The authors argue that entropy-TOPSIS is particularly appropriate when expert weight elicitation is logistically expensive at scale, and that the resulting rankings provide administrators with a basis to "make decisions scientifically, objectively, and limit subjective factors."

2.3 Applicant Selection: Indonesian SMPN 11 Case

Bagi et al. (2020) applied a hybrid AHP–TOPSIS pipeline to the candidate selection problem at SMPN 11, a school admitting students into a high-achievement track. Five criteria were weighted via AHP: grade average, attendance, extracurricular involvement, attitude, and readiness. After matrix normalization (Equation 1) and identification of positive and negative ideal solutions (Equation 2), candidates were ranked by the closeness coefficient defined in Equation 3.

The headline result is one of explicit divergence between manual and DSS-based rankings. The candidate "Ranti H." was ranked first by the DSS ($C_i = 0.823$) but only second by the manual review committee, with a scoring gap of approximately 0.09. The authors argue that manual ranking systematically undervalues candidates whose strengths are spread across multiple criteria rather than concentrated on a single dominant one—precisely the population that a uniformly-weighted TOPSIS will identify and that committee debate tends to overlook. As they put it, "TOPSIS can assist the selection committee in selecting high-achieving students."

3 Student-Side Applications

3.1 Career Path Selection: Indian Graduate Survey

Gupta and Kulkarni (2022) surveyed 502 graduate students across multiple disciplines on post-graduate decisions, identifying four primary career paths: dropping out to prepare for competitive exams, dropping out to join a family business, pursuing higher education, or seeking employment. Of the 502 respondents, 49.6% opted to drop out (across the two drop-out paths), and 32.3% pursued higher education. The authors report that TOPSIS rankings of the same career alternatives were "identical" to the rankings implied by the survey's empirical choice distribution—a concordance the source frames as evidence of reliability. The authors interpret this as evidence that "decisions are based on an individual's ability to take risk and rationale applied to opt for selecting an alternative," and that TOPSIS can capture both the internal motivations and the external pressures driving career choice.

3.2 University–Course Admission Choice: Nigerian Case

Oladokun and Oyewole (2015) examined a different student-side problem: matching student ability and competitiveness to specific course-and-university combinations. Rather than TOPSIS, the authors used a Mamdani fuzzy inference system that takes student ability and course competitiveness as inputs and produces a viability score for each student-course-university match. The system encodes a knowledge base in which, for example, "a high ability student applying to a highly competitive course and university will have a lower viability" than the same student applying to a moderately competitive course and university—a non-monotonic interaction that linear scoring rules cannot easily express. The authors argue that the linguistic, intuitive nature of the rule base makes it more accessible to admissions officers than the numerical Euclidean-distance machinery of TOPSIS, while still producing rankings that satisfy stakeholders. The Nigerian study is included here as the principal counterexample in the review: a setting where TOPSIS alone is judged insufficient and a fuzzy-rule-based DSS is preferred instead.

4 Handling Uncertainty: Fuzzy TOPSIS

4.1 Motivation

Linguistic data—ratings of "high," "medium," "low"—and imprecise numerical data are pervasive in admissions decisions and are not naturally handled by classical TOPSIS, which assumes crisp numerical inputs. Fuzzy TOPSIS (FTOPSIS) extends TOPSIS by replacing crisp scores with fuzzy numbers, most commonly Triangular Fuzzy Numbers (TFNs). Bhatia et al. (2024) apply FTOPSIS with a divergence measure to a university-selection problem and argue that "Fuzzy TOPSIS helps make accurate decisions regardless of the multiple criteria available," particularly when criteria are heterogeneous in measurement scale (continuous, ordinal, linguistic).

4.2 Triangular Fuzzy Numbers

A TFN $\tilde{p} = (g, h, k)$ with $g < h < k$ has membership function

$$u_{\tilde{p}}(x) = \begin{cases} \dfrac{x-g}{h-g}, & g \leq x \leq h \\ \dfrac{k-x}{k-h}, & h \leq x \leq k \\ 0, & x < g \text{ or } x > k \end{cases} \tag{4}$$

The lower limit g marks where membership begins to rise from zero, the peak h is the most-likely value (membership $= 1$), and the upper limit k marks where membership returns to zero (Figure 1). A 7-point Likert scale ranging from "Very Low" to "Very High" is typically mapped to seven TFNs whose peaks tile the $[0, 1]$ interval and whose supports overlap to encode the smoothness of linguistic categories.

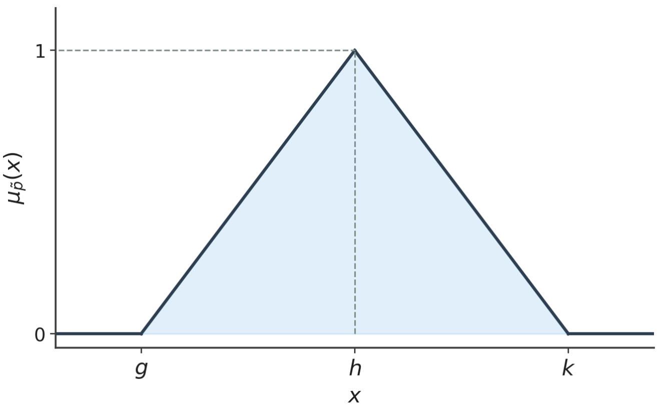

Figure 1: Membership function of a triangular fuzzy number $\tilde{p} = (g, h, k)$. Membership rises linearly from zero at g to one at h, then falls linearly back to zero at k.

4.3 Indian Higher Education Institutes Case

Koltharkar et al. (2020) applied FTOPSIS to prioritize student requirements at Higher Education Institutes (HEIs) in India, using an 8-dimension, 40-sub-metric instrument that captured students' priorities across academic, infrastructural, and social dimensions. Questionnaire responses on a 7-point Likert scale were converted into TFNs, weighted, and ranked. The methodological contribution is the explicit conversion pipeline from linguistic Likert response to TFN to weighted closeness coefficient—a pipeline that can be applied to any institutional self-assessment whose inputs are surveys rather than hard counts. The authors frame the resulting rankings as a tool to help HEIs "bear the critical responsibility of transforming the careers of their students" by directing infrastructural investment toward the dimensions students themselves identify as most lacking.

5 Discussion

The case studies surveyed converge on three claims. First, TOPSIS-based DSS identify ranking gaps that manual processes miss, especially for candidates whose strengths are distributed across multiple criteria rather than concentrated on a single dimension—the SMPN 11 case in Section 2.3 is the clearest example. Second, the AHP–TOPSIS pairing is mature: AHP handles the elicitation of subjective criterion weights, TOPSIS handles the ranking once weights are fixed, and entropy-TOPSIS provides a data-driven alternative for settings where expert elicitation is impractical. Third, FTOPSIS extends the framework to linguistic and imprecise data without requiring users to discard their natural rating vocabulary.

A useful contrast emerges between the Indonesian and Nigerian case studies. In the SMPN 11 setting (Bagi et al., 2020), where criteria are well-defined and numerically measurable (grade average, attendance), TOPSIS performed well and produced rankings that diverged from manual review in a direction the source frames as a correction. In the

Nigerian setting (Oladokun and Oyewole, 2015), where the matching of student ability to course competitiveness is non-monotonic and inherently linguistic, a fuzzy inference system was preferred over TOPSIS. This is consistent with a broader pattern in MCDM: TOPSIS is strongest when criteria are independent and reasonably numerical, and weakest when criteria interact in non-monotonic ways or are dominantly linguistic. FTOPSIS partially closes this gap, but does not fully recover the expressiveness of rule-based fuzzy inference.

The applications also differ in stake. Institutional self-evaluation (Sections 2.1 and 2.2) is comparatively low-risk: a misranked university affects accreditation and rankings but does not directly harm an individual. Applicant selection (Section 2.3) and career-path recommendation (Section 3.1) are higher-risk, because misrankings can affect individual life trajectories. None of the case studies summarized in this review describe sensitivity analyses for how robust their TOPSIS rankings are to small perturbations in criterion weights. This is an important gap for high-stakes deployment: a candidate whose rank depends on a small weight perturbation should arguably be flagged for committee review rather than admitted or rejected on the DSS output alone.

6 Conclusion

TOPSIS, particularly when paired with AHP for weight elicitation or with fuzzy extensions for linguistic input, offers a comprehensive and reproducible approach to multi-criteria decision-making in college admissions. The reviewed case studies demonstrate that TOPSIS-based DSS reduce subjective bias in candidate ranking and scale to large applicant pools. Where studies compared TOPSIS rankings directly against manual committee review (most clearly in the SMPN 11 case), the divergences are framed by the source literature as corrections rather than errors. Fuzzy TOPSIS further extends these benefits to linguistic and imprecise input data, which is where most real admissions data lives.

Three avenues of future work follow naturally. First, integration with machine learning—particularly for automated weight elicitation from historical admission outcomes—would reduce the subjectivity that AHP introduces at the weighting stage. Second, sensitivity analyses for how robust rankings are to weight perturbations would make TOPSIS deployable in high-stakes settings where defensibility matters. Third, none of the reviewed studies tracks downstream outcomes for admitted students. Linking DSS rankings to graduation, time-to-degree, and post-graduation employment would convert the existing literature, which validates TOPSIS against committee judgment, into one that validates TOPSIS against student outcomes.

For institutions weighing whether to deploy a TOPSIS-based DSS today, the reviewed evidence supports a hybrid approach: use the DSS as a structured first pass that ranks the full applicant pool and flags borderline cases for committee review, rather than as a replacement for committee judgment. This preserves the DSS's strengths in scale and consistency while retaining human review for the cases where rankings are most sensitive to weight choices. Thongdeelert et al. (2022) go further and argue that the DSS itself can become a feedback tool for advising staff, since "for guidance teachers, they can use the DSS as a tool to get the advice of their students"—turning the system from a one-shot ranking instrument into a longitudinal advising aid.

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